



**Syllabus:** Limits, Continuity, and Differentiability, Maxima and minima, Mean value theorem, Integration.

**Mark Distribution in Previous GATE**

| Year          | 2026 - 1 | 2026 - 2 | 2025 - 1 | 2025 - 2 | 2024 - 1 | 2024 - 2 | 2023 | 2022 | 2021 - 1 | 2021 - 2 | Minimum |
|---------------|----------|----------|----------|----------|----------|----------|------|------|----------|----------|---------|
| 1 Mark Count  | 1        | 1        | 1        | 1        | 1        | 1        | 2    | 1    | 1        | 1        | 1       |
| 2 Marks Count | 1        | 1        | 0        | 0        | 0        | 0        | 0    | 0    | 0        | 0        | 0       |
| Total Marks   | 3        | 3        | 1        | 1        | 1        | 1        | 2    | 1    | 1        | 1        | 1       |

Welcome to the "Engineering Mathematics: Calculus" chapter, a cornerstone of your GATE Computer Science preparation. This section is designed as a comprehensive, exam-focused reference to equip you with the essential concepts, formulas, and problem-solving strategies required to excel in the calculus portion of the GATE exam. Calculus is not merely a collection of mathematical techniques; it is the language of change and optimization, fundamental to understanding algorithms, data structures, probability, and various computational models. A solid grasp of calculus concepts will not only help you score well in the dedicated Engineering Mathematics section but also enhance your analytical abilities for other subjects. Typically, Engineering Mathematics accounts for 8-10% of the total GATE marks, with Calculus being a significant component of this weightage. Questions usually range from direct application of formulas and theorems to conceptual problems involving limits, continuity, differentiation, and integration, often requiring a blend of analytical reasoning and careful calculation. Expect a mix of Multiple Choice Questions (MCQ), Multiple Select Questions (MSQ), and Numerical Answer Type (NAT) questions.

## Topic-wise Key Concepts

### Limits

**Definition and Core Idea:** A limit describes the behavior of a function as its input approaches a certain value. It tells us what value the function 'tends to' or 'approaches' as the independent variable gets arbitrarily close to a specific point, without necessarily reaching it.

### Important Formulas, Theorems, and Results:

- Algebra of Limits:** If  $\lim_{x \rightarrow a} f(x) = L$  and  $\lim_{x \rightarrow a} g(x) = M$ , then:
  - $\lim_{x \rightarrow a} [f(x) \pm g(x)] = L \pm M$
  - $\lim_{x \rightarrow a} [f(x) \cdot g(x)] = L \cdot M$
  - $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{L}{M}$ , provided  $M \neq 0$
  - $\lim_{x \rightarrow a} [c \cdot f(x)] = c \cdot L$
- Standard Limits:**
  - $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$
  - $\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$
  - $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$
  - $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$
  - $\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \ln a$
  - $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1$
  - $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1}$
  - $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e$
  - $\lim_{x \rightarrow 0} (1+x)^{1/x} = e$
  - $\lim_{x \rightarrow \infty} \left(1 + \frac{k}{x}\right)^x = e^k$
- L'Hopital's Rule:** If  $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$  is of the indeterminate form  $\frac{0}{0}$  or  $\frac{\infty}{\infty}$ , then  $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$ , provided the latter limit exists. This can be applied repeatedly.
- Squeeze Theorem (Sandwich Theorem):** If  $g(x) \leq f(x) \leq h(x)$  for all  $x$  in an open interval containing  $c$  (except possibly at  $c$  itself), and if  $\lim_{x \rightarrow c} g(x) = L$  and  $\lim_{x \rightarrow c} h(x) = L$ , then  $\lim_{x \rightarrow c} f(x) = L$ .

### Key Properties and Identities:

- Indeterminate forms:  $\frac{0}{0}, \frac{\infty}{\infty}, 0 \cdot \infty, \infty - \infty, 1^\infty, 0^0, \infty^0$ . These require special techniques like L'Hopital's Rule or algebraic manipulation.

- One-sided limits: For a limit to exist, the left-hand limit ( $\lim_{x \rightarrow a^-} f(x)$ ) and the right-hand limit ( $\lim_{x \rightarrow a^+} f(x)$ ) must be equal.

### Common Pitfalls or Tricky Points:

- Applying L'Hopital's Rule when the limit is not an indeterminate form.
- Incorrectly handling one-sided limits, especially for piecewise functions or functions with square roots.
- Errors in algebraic simplification before applying standard limits.
- Misinterpreting limits involving infinity (e.g.,  $\frac{1}{\infty} = 0$ , but  $\infty \cdot 0$  is indeterminate).

### Standard Problem-Solving Techniques or Shortcuts:

- **Direct Substitution:** Always try this first. If it yields a finite value, that's the limit.
- **Factorization and Cancellation:** For  $\frac{0}{0}$  forms involving polynomials, factorize the numerator and denominator to cancel common terms.
- **Rationalization:** For expressions involving square roots, multiply by the conjugate to simplify.
- **Using Standard Limits:** Manipulate the expression to fit one of the standard limit forms.
- **L'Hopital's Rule:** Apply for  $\frac{0}{0}$  or  $\frac{\infty}{\infty}$  forms. Remember to differentiate numerator and denominator separately.
- **Transforming Indeterminate Forms:**
  - $0 \cdot \infty \implies \frac{0}{1/\infty}$  or  $\frac{\infty}{1/0}$
  - $\infty - \infty \implies$  combine terms, rationalize, or find a common denominator.
  - $1^\infty, 0^0, \infty^0 \implies$  take natural logarithm: If  $y = [f(x)]^{g(x)}$ , then  $\ln y = g(x) \ln f(x)$ . Then find  $\lim \ln y$  and exponentiate the result.

### Continuity

**Definition and Core Idea:** A function is continuous at a point if its graph has no breaks, jumps, or holes at that point. Informally, you can draw the graph without lifting your pen. Formally, it means the function's value at the point is equal to its limit as the input approaches that point.

### Important Formulas, Theorems, and Results:

1. **Conditions for Continuity at a Point  $x = a$ :** A function  $f(x)$  is continuous at  $x = a$  if and only if all three conditions are met:
  - a.  $f(a)$  is defined (the function exists at  $a$ ).
  - b.  $\lim_{x \rightarrow a} f(x)$  exists (the limit exists at  $a$ ). This implies  $\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x)$ .
  - c.  $\lim_{x \rightarrow a} f(x) = f(a)$  (the limit equals the function value).
2. **Continuity on an Interval:** A function is continuous on an open interval  $(a, b)$  if it is continuous at every point in the interval. It is continuous on a closed interval  $[a, b]$  if it is continuous on  $(a, b)$ , continuous from the right at  $a$  ( $\lim_{x \rightarrow a^+} f(x) = f(a)$ ), and continuous from the left at  $b$  ( $\lim_{x \rightarrow b^-} f(x) = f(b)$ ).
3. **Intermediate Value Theorem (IVT):** If  $f$  is continuous on the closed interval  $[a, b]$  and  $k$  is any number between  $f(a)$  and  $f(b)$ , then there exists at least one number  $c$  in  $(a, b)$  such that  $f(c) = k$ .
4. **Extreme Value Theorem (EVT):** If  $f$  is continuous on a closed interval  $[a, b]$ , then  $f$  attains both an absolute maximum value and an absolute minimum value on that interval.

### Key Properties and Identities:

- The sum, difference, product, and quotient (where the denominator is non-zero) of continuous functions are continuous.
- The composition of continuous functions is continuous: If  $g$  is continuous at  $a$  and  $f$  is continuous at  $g(a)$ , then  $(f \circ g)(x) = f(g(x))$  is continuous at  $a$ .
- Polynomials, exponential functions ( $e^x, a^x$ ), logarithmic functions ( $\ln x, \log_a x$ ), trigonometric functions ( $\sin x, \cos x$ ), and rational functions (where the denominator is non-zero) are continuous in their domains.

### Common Pitfalls or Tricky Points:

- **Piecewise Functions:** Pay close attention to the points where the definition of the function changes. These are the critical points to check for continuity.
- **Removable Discontinuities:** Occur when  $\lim_{x \rightarrow a} f(x)$  exists but is not equal to  $f(a)$ , or  $f(a)$  is undefined (e.g., holes in the graph).
- **Jump Discontinuities:** Occur when the left-hand and right-hand limits exist but are not equal (e.g., step functions).

- **Infinite Discontinuities:** Occur when one or both of the one-sided limits are infinite (e.g., vertical asymptotes).
- Forgetting to check all three conditions for continuity.

### Standard Problem-Solving Techniques or Shortcuts:

- For piecewise functions, evaluate the function value at the transition point, and calculate both the left-hand and right-hand limits at that point. Equate them to find unknown constants.
- Identify the domain of the function first. Discontinuities often occur at points where the function is undefined (e.g., denominator is zero, negative under square root).
- For functions involving absolute values, rewrite them as piecewise functions.

### Differentiation

**Definition and Core Idea:** Differentiation is the process of finding the derivative of a function. The derivative represents the instantaneous rate of change of a function with respect to its independent variable. Geometrically, it gives the slope of the tangent line to the function's graph at a given point.

### Important Formulas, Theorems, and Results:

1. **Definition of Derivative:**  $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

2. **Basic Differentiation Rules:**

- $\frac{d}{dx}(c) = 0$  (where  $c$  is a constant)
- $\frac{d}{dx}(x^n) = nx^{n-1}$
- $\frac{d}{dx}(e^x) = e^x$
- $\frac{d}{dx}(a^x) = a^x \ln a$
- $\frac{d}{dx}(\ln x) = \frac{1}{x}$
- $\frac{d}{dx}(\log_a x) = \frac{1}{x \ln a}$
- $\frac{d}{dx}(\sin x) = \cos x$
- $\frac{d}{dx}(\cos x) = -\sin x$
- $\frac{d}{dx}(\tan x) = \sec^2 x$
- $\frac{d}{dx}(\cot x) = -\csc^2 x$
- $\frac{d}{dx}(\sec x) = \sec x \tan x$
- $\frac{d}{dx}(\csc x) = -\csc x \cot x$
- $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$
- $\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$
- $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$

3. **Rules of Differentiation:**

- **Constant Multiple Rule:**  $\frac{d}{dx}[cf(x)] = c \frac{d}{dx}[f(x)]$
- **Sum/Difference Rule:**  $\frac{d}{dx}[f(x) \pm g(x)] = \frac{d}{dx}[f(x)] \pm \frac{d}{dx}[g(x)]$
- **Product Rule:**  $\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x)$
- **Quotient Rule:**  $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$
- **Chain Rule:**  $\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$

4. **Implicit Differentiation:** Used when  $y$  is not explicitly defined as a function of  $x$ . Differentiate both sides of the equation with respect to  $x$ , treating  $y$  as a function of  $x$  (i.e., use the chain rule for terms involving  $y$ ).

5. **Rolle's Theorem:** If  $f$  is continuous on  $[a, b]$ , differentiable on  $(a, b)$ , and  $f(a) = f(b)$ , then there exists at least one  $c \in (a, b)$  such that  $f'(c) = 0$ .

6. **Mean Value Theorem (Lagrange's MVT):** If  $f$  is continuous on  $[a, b]$  and differentiable on  $(a, b)$ , then there exists at least one  $c \in (a, b)$  such that  $f'(c) = \frac{f(b) - f(a)}{b - a}$ .

7. **Cauchy's Mean Value Theorem:** If  $f$  and  $g$  are continuous on  $[a, b]$  and differentiable on  $(a, b)$ , and  $g'(x) \neq 0$  for any  $x \in (a, b)$ , then there exists at least one  $c \in (a, b)$  such that  $\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$ .

### Key Properties and Identities:

- Differentiability implies continuity, but continuity does not imply differentiability (e.g.,  $|x|$  at  $x = 0$ ).

- Higher-order derivatives:  $f''(x) = \frac{d}{dx}(f'(x))$ ,  $f'''(x) = \frac{d}{dx}(f''(x))$ , etc.

### Common Pitfalls or Tricky Points:

- Forgetting the chain rule, especially with composite functions like  $\sin(2x)$  or  $e^{x^2}$ .
- Errors in applying the product or quotient rule.
- Mistakes in implicit differentiation, particularly when differentiating terms involving  $y$  (e.g.,  $\frac{d}{dx}(y^2) = 2y\frac{dy}{dx}$ ).
- Confusing the derivative of  $a^x$  with  $x^a$ .
- Not simplifying expressions before differentiating when possible.

### Standard Problem-Solving Techniques or Shortcuts:

- Logarithmic Differentiation:** For functions of the form  $y = [f(x)]^{g(x)}$  or complex products/quotients, take the natural logarithm of both sides before differentiating implicitly.
- Parametric Differentiation:** If  $x = f(t)$  and  $y = g(t)$ , then  $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$ .
- Derivative of Inverse Function:** If  $y = f^{-1}(x)$ , then  $\frac{dy}{dx} = \frac{1}{f'(y)}$ .
- Practice recognizing common derivative patterns for speed.

### Maxima Minima

**Definition and Core Idea:** Maxima and minima (collectively called extrema) refer to the highest and lowest values a function attains within a given interval or its entire domain. A local maximum/minimum is the highest/lowest point in its immediate neighborhood, while a global (absolute) maximum/minimum is the highest/lowest point over the entire domain/interval.

### Important Formulas, Theorems, and Results:

- Critical Points:** A critical point of a function  $f(x)$  is a point  $c$  in the domain of  $f$  where either  $f'(c) = 0$  or  $f'(c)$  is undefined. Local extrema can only occur at critical points or endpoints of a closed interval.
- First Derivative Test:** Let  $c$  be a critical point of a continuous function  $f$ .
  - If  $f'(x)$  changes from positive to negative at  $c$ , then  $f$  has a local maximum at  $c$ .
  - If  $f'(x)$  changes from negative to positive at  $c$ , then  $f$  has a local minimum at  $c$ .
  - If  $f'(x)$  does not change sign at  $c$ , then  $f$  has neither a local maximum nor a local minimum at  $c$ .
- Second Derivative Test:** Let  $c$  be a critical point where  $f'(c) = 0$  and  $f''(c)$  exists.
  - If  $f''(c) > 0$ , then  $f$  has a local minimum at  $c$ .
  - If  $f''(c) < 0$ , then  $f$  has a local maximum at  $c$ .
  - If  $f''(c) = 0$ , the test is inconclusive (use the First Derivative Test).
- Concavity:**
  - If  $f''(x) > 0$  on an interval,  $f$  is concave up (graph opens upwards).
  - If  $f''(x) < 0$  on an interval,  $f$  is concave down (graph opens downwards).
- Inflection Point:** A point where the concavity of the function changes. This occurs where  $f''(x) = 0$  or  $f''(x)$  is undefined, and  $f''(x)$  changes sign.

### Key Properties and Identities:

- For global extrema on a closed interval  $[a, b]$ , evaluate  $f(x)$  at all critical points in  $(a, b)$  and at the endpoints  $a$  and  $b$ . The largest of these values is the absolute maximum, and the smallest is the absolute minimum.

### Common Pitfalls or Tricky Points:

- Forgetting to check endpoints when finding global extrema on a closed interval.
- Misinterpreting the results of the second derivative test, especially when  $f''(c) = 0$ .
- Not considering points where  $f'(x)$  is undefined as critical points.
- Errors in algebraic manipulation while finding critical points or second derivatives.
- Confusing local extrema with global extrema.

### Standard Problem-Solving Techniques or Shortcuts:

- Optimization Problems:**
  - Understand the problem and identify the quantity to be maximized or minimized.

2. Draw a diagram if helpful.
  3. Introduce variables and write the primary equation for the quantity to be optimized.
  4. Write a secondary equation (constraint) relating the variables.
  5. Use the secondary equation to express the primary equation in terms of a single independent variable.
  6. Find the domain of this function.
  7. Differentiate the function, find critical points, and apply the First or Second Derivative Test.
  8. Verify the solution makes sense in the context of the problem.
- For simple functions, sometimes the graph can quickly indicate extrema.

## Integration

**Definition and Core Idea:** Integration is the reverse process of differentiation, also known as antidifferentiation. It finds the function whose derivative is the given function. Geometrically, definite integration calculates the area under the curve of a function between two specified limits.

### Important Formulas, Theorems, and Results:

#### 1. Basic Integration Formulas (Indefinite Integrals):

- $\int k \, dx = kx + C$
- $\int x^n \, dx = \frac{x^{n+1}}{n+1} + C$ , for  $n \neq -1$
- $\int \frac{1}{x} \, dx = \ln|x| + C$
- $\int e^x \, dx = e^x + C$
- $\int a^x \, dx = \frac{a^x}{\ln a} + C$
- $\int \sin x \, dx = -\cos x + C$
- $\int \cos x \, dx = \sin x + C$
- $\int \sec^2 x \, dx = \tan x + C$
- $\int \csc^2 x \, dx = -\cot x + C$
- $\int \sec x \tan x \, dx = \sec x + C$
- $\int \csc x \cot x \, dx = -\csc x + C$
- $\int \frac{1}{\sqrt{a^2 - x^2}} \, dx = \sin^{-1}\left(\frac{x}{a}\right) + C$
- $\int \frac{1}{a^2 + x^2} \, dx = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right) + C$
- $\int \frac{1}{x\sqrt{x^2 - a^2}} \, dx = \frac{1}{a} \sec^{-1}\left(\frac{x}{a}\right) + C$

#### 2. Properties of Indefinite Integrals:

- $\int [f(x) \pm g(x)] \, dx = \int f(x) \, dx \pm \int g(x) \, dx$
- $\int c f(x) \, dx = c \int f(x) \, dx$

#### 3. Integration by Substitution (u-substitution):

If  $u = g(x)$ , then  $du = g'(x) \, dx$ .  $\int f(g(x))g'(x) \, dx = \int f(u) \, du$ .

#### 4. Integration by Parts:

$\int u \, dv = uv - \int v \, du$ . Often remembered by the mnemonic "LIATE" (Logarithmic, Inverse trig, Algebraic, Trigonometric, Exponential) for choosing  $u$ .

### Key Properties and Identities:

- Always include the constant of integration  $+C$  for indefinite integrals.
- Trigonometric identities are often crucial for simplifying integrands before integration.

### Common Pitfalls or Tricky Points:

- Forgetting the constant of integration  $+C$ .
- Incorrectly choosing  $u$  and  $dv$  in integration by parts.
- Errors in algebraic manipulation or trigonometric identities.
- Not recognizing when substitution is appropriate.
- Confusing integration formulas with differentiation formulas (e.g.,  $\int \frac{1}{x} \, dx$  vs.  $\frac{d}{dx} \ln x$ ).

### Standard Problem-Solving Techniques or Shortcuts:

- **Direct Integration:** If the integrand matches a basic formula.
- **Substitution:** Look for a function and its derivative within the integrand. Let the inner function be  $u$ .
- **Integration by Parts:** Use for products of functions (e.g.,  $xe^x$ ,  $x \sin x$ ,  $\ln x$ ).
- **Partial Fractions:** For rational functions (polynomials divided by polynomials), decompose into simpler fractions that are easier to integrate. (Briefly mention as it's a common technique).

- **Trigonometric Identities:** Use identities to simplify powers of trig functions or products of trig functions.

## Definite Integral

**Definition and Core Idea:** A definite integral represents the net signed area between the graph of a function and the x-axis over a specified interval  $[a, b]$ . It is evaluated by finding the antiderivative of the function and then evaluating it at the upper and lower limits of integration.

### Important Formulas, Theorems, and Results:

1. **Fundamental Theorem of Calculus (Part 1):** If  $f$  is continuous on  $[a, b]$ , then the function  $G(x) = \int_a^x f(t) dt$  has a derivative at every point in  $(a, b)$ , and  $G'(x) = f(x)$ .
2. **Fundamental Theorem of Calculus (Part 2):** If  $f$  is continuous on  $[a, b]$  and  $F$  is any antiderivative of  $f$  on  $[a, b]$ , then  $\int_a^b f(x) dx = F(b) - F(a)$ .
3. **Properties of Definite Integrals:**
  - $\int_a^b f(x) dx = -\int_b^a f(x) dx$
  - $\int_a^a f(x) dx = 0$
  - $\int_a^b cf(x) dx = c \int_a^b f(x) dx$
  - $\int_a^b [f(x) \pm g(x)] dx = \int_a^b f(x) dx \pm \int_a^b g(x) dx$
  - $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$ , where  $a < c < b$
  - $\int_0^a f(x) dx = \int_0^a f(a-x) dx$  (useful for simplifying integrals)
  - **Even/Odd Functions:**
    - If  $f(x)$  is an even function ( $f(-x) = f(x)$ ), then  $\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$ .
    - If  $f(x)$  is an odd function ( $f(-x) = -f(x)$ ), then  $\int_{-a}^a f(x) dx = 0$ .

### Key Properties and Identities:

- The definite integral evaluates to a numerical value, not a function.
- The constant of integration  $+C$  is not included in definite integrals because it cancels out ( $F(b) + C - (F(a) + C) = F(b) - F(a)$ ).

### Common Pitfalls or Tricky Points:

- Incorrectly applying the limits of integration (upper limit minus lower limit).
- Sign errors when evaluating  $F(b) - F(a)$ .
- Forgetting to change the limits of integration when using substitution for definite integrals.
- Not recognizing even/odd function properties to simplify integrals over symmetric intervals.
- Improper integrals (involving infinite limits or discontinuities within the interval) are less common in GATE CS but good to be aware of.

### Standard Problem-Solving Techniques or Shortcuts:

- **Direct Application of FTC:** Find the antiderivative and evaluate at the limits.
- **Substitution:** If using substitution, either change the limits of integration to be in terms of  $u$  or revert back to  $x$  before applying the original limits.
- **Properties of Definite Integrals:** Use properties like linearity, interval splitting, and even/odd function rules to simplify the integral before evaluation.
- **Symmetry:** For integrals over  $[-a, a]$ , check if the integrand is even or odd.

## Polynomials

**Definition and Core Idea:** A polynomial is an expression consisting of variables and coefficients, involving only the operations of addition, subtraction, multiplication, and non-negative integer exponents of variables. They are fundamental building blocks in calculus as they are continuous and differentiable everywhere.

### Important Formulas, Theorems, and Results:

1. **General Form:**  $P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$ , where  $a_i$  are coefficients and  $n$  is a non-negative integer (degree).
2. **Remainder Theorem:** If a polynomial  $P(x)$  is divided by  $(x - a)$ , the remainder is  $P(a)$ .

3. **Factor Theorem:**  $(x - a)$  is a factor of a polynomial  $P(x)$  if and only if  $P(a) = 0$  (i.e.,  $a$  is a root of the polynomial).
4. **Fundamental Theorem of Algebra:** Every non-constant single-variable polynomial with complex coefficients has at least one complex root. Consequently, a polynomial of degree  $n$  has exactly  $n$  roots (counting multiplicity) in the complex numbers.
5. **Vieta's Formulas (for quadratic and cubic):**
  - For  $ax^2 + bx + c = 0$  with roots  $\alpha, \beta$ :
    - $\alpha + \beta = -\frac{b}{a}$
    - $\alpha\beta = \frac{c}{a}$
  - For  $ax^3 + bx^2 + cx + d = 0$  with roots  $\alpha, \beta, \gamma$ :
    - $\alpha + \beta + \gamma = -\frac{b}{a}$
    - $\alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}$
    - $\alpha\beta\gamma = -\frac{d}{a}$

#### Key Properties and Identities:

- Polynomials are continuous and infinitely differentiable everywhere.
- The degree of a polynomial determines the maximum number of roots it can have.
- If a polynomial has real coefficients, then any complex roots must occur in conjugate pairs.

#### Common Pitfalls or Tricky Points:

- Errors in polynomial division or synthetic division.
- Misapplying Vieta's formulas (e.g., sign errors).
- Not considering multiplicity of roots.
- Confusing roots with factors.

#### Standard Problem-Solving Techniques or Shortcuts:

- **Synthetic Division:** A quick method for dividing a polynomial by a linear factor  $(x - a)$ .
- **Rational Root Theorem:** Helps find potential rational roots of a polynomial with integer coefficients.
- **Factoring:** Look for common factors, use grouping, or quadratic formula for quadratic factors.
- **Derivative for Repeated Roots:** If  $a$  is a root of  $P(x)$  with multiplicity  $k > 1$ , then  $a$  is also a root of  $P'(x)$  with multiplicity  $k - 1$ .

#### Summation

**Definition and Core Idea:** Summation is the process of adding a sequence of numbers or terms. It is often represented using sigma notation ( $\sum$ ). In calculus, summation is foundational to understanding series, Riemann sums (for integration), and numerical methods.

#### Important Formulas, Theorems, and Results:

1. **Arithmetic Progression (AP) Sum:** For an AP with first term  $a$ , common difference  $d$ , and  $n$  terms:
  - $S_n = \frac{n}{2}[2a + (n - 1)d]$
  - $S_n = \frac{n}{2}[a + l]$ , where  $l$  is the last term.
2. **Geometric Progression (GP) Sum:** For a GP with first term  $a$ , common ratio  $r$ , and  $n$  terms:
  - $S_n = \frac{a(r^n - 1)}{r - 1}$ , for  $r \neq 1$
  - $S_n = \frac{a(1 - r^n)}{1 - r}$ , for  $r \neq 1$
  - **Sum to Infinity for GP:**  $S_\infty = \frac{a}{1 - r}$ , for  $|r| < 1$
3. **Standard Summation Formulas:**
  - $\sum_{i=1}^n c = nc$
  - $\sum_{i=1}^n i = \frac{n(n+1)}{2}$
  - $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$
  - $\sum_{i=1}^n i^3 = \left(\frac{n(n+1)}{2}\right)^2$

#### Key Properties and Identities:

- **Linearity of Summation:**

- $\sum_{i=1}^n (a_i \pm b_i) = \sum_{i=1}^n a_i \pm \sum_{i=1}^n b_i$
- $\sum_{i=1}^n c \cdot a_i = c \cdot \sum_{i=1}^n a_i$
- **Telescoping Sums:** Sums where intermediate terms cancel out, e.g.,  $\sum_{i=1}^n (a_i - a_{i+1})$ .

### Common Pitfalls or Tricky Points:

- Off-by-one errors in summation limits.
- Confusing AP and GP formulas.
- Incorrectly applying the sum to infinity formula (only valid for  $|r| < 1$ ).
- Errors in algebraic manipulation when simplifying complex summations.

### Standard Problem-Solving Techniques or Shortcuts:

- **Identify Series Type:** Determine if it's an AP, GP, or a combination.
- **Use Standard Formulas:** Apply the formulas for  $\sum i$ ,  $\sum i^2$ ,  $\sum i^3$ .
- **Split and Combine:** Use linearity to break down complex summations into simpler ones.
- **Method of Differences (Telescoping Sums):** Express each term as a difference of two consecutive terms to simplify the sum.
- **Generating Functions:** (Advanced, less common for direct GATE CS questions, but good for understanding series).

## Quick Formula Reference

### Limits

- $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$
- $\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$
- $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$
- $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$
- $\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \ln a$
- $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1$
- $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1}$
- $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e$
- $\lim_{x \rightarrow 0} (1 + x)^{1/x} = e$
- L'Hopital's Rule: If  $\frac{f(x)}{g(x)}$  is  $\frac{0}{0}$  or  $\frac{\infty}{\infty}$ , then  $\lim \frac{f(x)}{g(x)} = \lim \frac{f'(x)}{g'(x)}$ .

### Continuity

- Continuous at  $x = a$  if:  $f(a)$  defined,  $\lim_{x \rightarrow a} f(x)$  exists, and  $\lim_{x \rightarrow a} f(x) = f(a)$ .

### Differentiation

Let  $u, v$  be functions of  $x$ ,  $c$  be a constant.

- $\frac{d}{dx}(c) = 0$
- $\frac{d}{dx}(x^n) = nx^{n-1}$
- $\frac{d}{dx}(e^x) = e^x$
- $\frac{d}{dx}(a^x) = a^x \ln a$
- $\frac{d}{dx}(\ln x) = \frac{1}{x}$
- $\frac{d}{dx}(\sin x) = \cos x$
- $\frac{d}{dx}(\cos x) = -\sin x$
- $\frac{d}{dx}(\tan x) = \sec^2 x$
- $\frac{d}{dx}(cu) = c \frac{du}{dx}$
- $\frac{d}{dx}(u \pm v) = \frac{du}{dx} \pm \frac{dv}{dx}$
- $\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$

- $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$
- $\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$  (Chain Rule)
- Rolle's Theorem:  $f(a) = f(b) \implies \exists c \in (a, b)$  s.t.  $f'(c) = 0$
- Mean Value Theorem:  $\exists c \in (a, b)$  s.t.  $f'(c) = \frac{f(b) - f(a)}{b - a}$

## Maxima Minima

- Critical points:  $f'(x) = 0$  or  $f'(x)$  undefined.
- First Derivative Test: Sign change of  $f'(x)$  at critical point.
- Second Derivative Test: If  $f'(c) = 0$ :  $f''(c) > 0 \implies$  local min;  $f''(c) < 0 \implies$  local max.

## Integration (Indefinite)

- $\int k dx = kx + C$
- $\int x^n dx = \frac{x^{n+1}}{n+1} + C$ , for  $n \neq -1$
- $\int \frac{1}{x} dx = \ln|x| + C$
- $\int e^x dx = e^x + C$
- $\int a^x dx = \frac{a^x}{\ln a} + C$
- $\int \sin x dx = -\cos x + C$
- $\int \cos x dx = \sin x + C$
- $\int \sec^2 x dx = \tan x + C$
- $\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1}\left(\frac{x}{a}\right) + C$
- $\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right) + C$
- Integration by Parts:  $\int u dv = uv - \int v du$

## Definite Integral

- Fundamental Theorem of Calculus:  $\int_a^b f(x) dx = F(b) - F(a)$  where  $F'(x) = f(x)$ .
- $\int_a^b f(x) dx = -\int_b^a f(x) dx$
- $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$
- $\int_0^a f(x) dx = \int_0^a f(a-x) dx$
- Even function ( $f(-x) = f(x)$ ):  $\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$
- Odd function ( $f(-x) = -f(x)$ ):  $\int_{-a}^a f(x) dx = 0$

## Polynomials

- Remainder Theorem:  $P(x)$  divided by  $(x - a)$  has remainder  $P(a)$ .
- Factor Theorem:  $(x - a)$  is a factor of  $P(x)$  iff  $P(a) = 0$ .
- Vieta's Formulas (for  $ax^2 + bx + c = 0$ ):  $\alpha + \beta = -b/a$ ,  $\alpha\beta = c/a$ .

## Summation

- AP Sum:  $S_n = \frac{n}{2}[2a + (n-1)d]$  or  $S_n = \frac{n}{2}[a + l]$
- GP Sum:  $S_n = \frac{a(r^n - 1)}{r - 1}$
- GP Sum to Infinity:  $S_\infty = \frac{a}{1-r}$  for  $|r| < 1$
- $\sum_{i=1}^n i = \frac{n(n+1)}{2}$
- $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$
- $\sum_{i=1}^n i^3 = \left(\frac{n(n+1)}{2}\right)^2$

## Important Tips for GATE

1. **Master the Fundamentals:** Calculus builds upon basic algebra, trigonometry, and functions. Ensure your foundation in these areas is strong to avoid silly mistakes.
2. **Practice with PYQs:** Solve previous year's GATE questions extensively. This helps you understand the typical question patterns, difficulty levels, and time constraints.
3. **Understand Concepts, Don't Just Memorize:** While formulas are crucial, knowing \*why\* they work (e.g., the geometric interpretation of derivatives and integrals) will help you tackle trickier conceptual questions and apply them correctly in unfamiliar scenarios.
4. **Pay Attention to Domains and Conditions:** For limits, continuity, and differentiability, always check the domain of the function and any specific conditions (e.g.,  $g(x) \neq 0$  for quotient rule,  $|r| < 1$  for infinite GP sum). Piecewise functions are common traps.
5. **Be Careful with Signs and Constants:** A common source of error in differentiation and integration is incorrect signs or forgetting the constant of integration ( $+C$ ) for indefinite integrals. Double-check your calculations.
6. **Time Management:** Some calculus problems can be lengthy. Learn to identify problems that might take too much time and mark them for review. Practice solving problems within a strict time limit. Shortcuts like L'Hopital's rule or properties of definite integrals can save valuable time.
7. **Visualize Functions:** For problems involving maxima/minima or definite integrals, sketching a rough graph of the function can often provide intuition and help verify your analytical results.
8. **Review Properties of Even/Odd Functions:** These properties are frequently tested in definite integral problems, especially over symmetric intervals like  $[-a, a]$ , and can significantly simplify calculations.

## 5.1 Continuity (11)

 **Practice Test:** Test 1 (14Q)

### 5.1.1 Continuity: GATE CSE 1996 | Question: 3



Let  $f$  be a function defined by

$$f(x) = \begin{cases} x^2 & \text{for } x \leq 1 \\ ax^2 + bx + c & \text{for } 1 < x \leq 2 \\ x + d & \text{for } x > 2 \end{cases}$$

Find the values for the constants  $a, b, c$  and  $d$  so that  $f$  is continuous and differentiable everywhere on the real line.

gate1996 calculus continuity differentiation normal descriptive

**Answer key** 

### 5.1.2 Continuity: GATE CSE 1998 | Question: 1.4



Consider the function  $y = |x|$  in the interval  $[-1, 1]$ . In this interval, the function is

- |                                      |                                          |
|--------------------------------------|------------------------------------------|
| A. continuous and differentiable     | B. continuous but not differentiable     |
| C. differentiable but not continuous | D. neither continuous nor differentiable |

gate1998 calculus continuity differentiation easy

**Answer key** 

### 5.1.3 Continuity: GATE CSE 2007 | Question: 1



Consider the following two statements about the function  $f(x) = |x|$ :

- P.  $f(x)$  is continuous for all real values of  $x$ .
- Q.  $f(x)$  is differentiable for all real values of  $x$ .

Which of the following is **TRUE**?

- |                                  |                                  |
|----------------------------------|----------------------------------|
| A. $P$ is true and $Q$ is false. | B. $P$ is false and $Q$ is true. |
| C. Both $P$ and $Q$ are true.    | D. Both $P$ and $Q$ are false.   |

gatecse-2007 calculus continuity differentiation easy

Answer key

5.1.4 Continuity: GATE CSE 2013 | Question: 22



Which one of the following functions is continuous at  $x = 3$ ?

- A.  $f(x) = \begin{cases} 2, & \text{if } x = 3 \\ x - 1, & \text{if } x > 3 \\ \frac{x+3}{3}, & \text{if } x < 3 \end{cases}$
- B.  $f(x) = \begin{cases} 4, & \text{if } x = 3 \\ 8 - x, & \text{if } x \neq 3 \end{cases}$
- C.  $f(x) = \begin{cases} x + 3, & \text{if } x \leq 3 \\ x - 4, & \text{if } x > 3 \end{cases}$
- D.  $f(x) = \begin{cases} \frac{1}{x^3 - 27}, & \text{if } x \neq 3 \end{cases}$

gatecse-2013 calculus continuity normal

Answer key

5.1.5 Continuity: GATE CSE 2014 | Set 1 | Question: 47



A function  $f(x)$  is continuous in the interval  $[0, 2]$ . It is known that  $f(0) = f(2) = -1$  and  $f(1) = 1$ . Which one of the following statements must be true?

- A. There exists a  $y$  in the interval  $(0, 1)$  such that  $f(y) = f(y + 1)$
- B. For every  $y$  in the interval  $(0, 1)$ ,  $f(y) = f(2 - y)$
- C. The maximum value of the function in the interval  $(0, 2)$  is 1
- D. There exists a  $y$  in the interval  $(0, 1)$  such that  $f(y) = -f(2 - y)$

gatecse-2014-set1 calculus continuity tricky

Answer key

5.1.6 Continuity: GATE CSE 2015 | Set 2 | Question: 26



Let  $f(x) = x^{-\frac{1}{3}}$  and  $A$  denote the area of region bounded by  $f(x)$  and the X-axis, when  $x$  varies from  $-1$  to  $1$ . Which of the following statements is/are TRUE?

- I.  $f$  is continuous in  $[-1, 1]$
- II.  $f$  is not bounded in  $[-1, 1]$
- III.  $A$  is nonzero and finite

- A. II only                      B. III only                      C. II and III only                      D. I, II and III

gatecse-2015-set2 calculus continuity functions normal

Answer key

5.1.7 Continuity: GATE CSE 2021 | Set 2 | Question: 25



Suppose that  $f : \mathbb{R} \rightarrow \mathbb{R}$  is a continuous function on the interval  $[-3, 3]$  and a differentiable function in the interval  $(-3, 3)$  such that for every  $x$  in the interval,  $f'(x) \leq 2$ . If  $f(-3) = 7$ , then  $f(3)$  is at most \_\_\_\_\_

gatecse-2021-set2 numerical-answers calculus continuity one-mark

Answer key

5.1.8 Continuity: GATE CSE 2026 | Set 1 | Question: 22



Consider the function  $f : \mathbb{R} \rightarrow \mathbb{R}$  defined as follows:

$$f(x) = \begin{cases} c_1 e^x - c_2 \log_e\left(\frac{1}{x}\right), & \text{if } x > 0 \\ 3 & \text{otherwise} \end{cases}$$

where  $c_1, c_2 \in \mathbb{R}$ .

If  $f$  is continuous at  $x = 0$ , then  $c_1 + c_2 = \underline{\hspace{2cm}}$ . (answer in integer)

gatecse-2026-set1 calculus continuity numerical-answers one-mark

Answer key

#### 5.1.9 Continuity: GATE CSE 2026 | Set 2 | Question: 54



Consider a function  $f : (0, 1) \rightarrow \{0, 1\}$  defined as follows.

For a real number  $r \in (0, 1)$ ,  $f(r) = 1$  if the second digit after the decimal point in  $r$  is one of the four digits 2, 3, 6 and 7. Otherwise,  $f(r)$  is equal to 0.

The number of points in  $(0, 1)$  at which  $f$  is discontinuous is         . (answer in integer)

gatecse-2026-set2 calculus continuity numerical-answers two-marks

Answer key

#### 5.1.10 Continuity: GATE DS&AI 2024 | Question: 27



Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be a function. Note:  $\mathbb{R}$  denotes the set of real numbers.

$$f(x) = \begin{cases} -x, & \text{if } x < -2 \\ ax^2 + bx + c, & \text{if } x \in [-2, 2] \\ x, & \text{if } x > 2 \end{cases}$$

Which ONE of the following choices gives the values of  $a, b, c$  that make the function  $f$  continuous and differentiable?

A.  $a = \frac{1}{4}, b = 0, c = 1$   
C.  $a = 0, b = 0, c = 0$

B.  $a = \frac{1}{2}, b = 0, c = 0$   
D.  $a = 1, b = 1, c = -4$

gate-ds-ai-2024 calculus continuity differentiation two-marks

Answer key

#### 5.1.11 Continuity: GATE2010 ME



The function  $y = |2 - 3x|$

- A. **is continuous**  $\forall x \in \mathbb{R}$  and differentiable  $\forall x \in \mathbb{R}$
- B. **is continuous**  $\forall x \in \mathbb{R}$  and differentiable  $\forall x \in \mathbb{R}$  except at  $x = \frac{3}{2}$
- C. **is continuous**  $\forall x \in \mathbb{R}$  and differentiable  $\forall x \in \mathbb{R}$  except at  $x = \frac{2}{3}$
- D. **is continuous**  $\forall x \in \mathbb{R}$  except  $x = 3$  and differentiable  $\forall x \in \mathbb{R}$

calculus gate2010me engineering-mathematics continuity

Answer key

## 5.2

## Definite Integral (4)

**Practice Test:** Test 1 (12Q)

### 5.2.1 Definite Integral: GATE CSE 2023 | Question: 21



The value of the definite integral

$$\int_{-3}^3 \int_{-2}^2 \int_{-1}^1 (4x^2y - z^3) dz dy dx$$

is \_\_\_\_\_. (Rounded off to the nearest integer)

gatecse-2023 calculus definite-integral numerical-answers one-mark

Answer key

### 5.2.2 Definite Integral: GATE CSE 2024 | Set 2 | Question: 6



Let  $f(x)$  be a continuous function from  $\mathbb{R}$  to  $\mathbb{R}$  such that

$$f(x) = 1 - f(2 - x)$$

Which one of the following options is the CORRECT value of  $\int_0^2 f(x) dx$  ?

- A. 0                                      B. 1                                      C. 2                                      D. -1

gatecse-2024-set2 calculus definite-integral one-mark

Answer key

### 5.2.3 Definite Integral: GATE CSE 2025 | Set 2 | Question: 2



The value of  $x$  such that  $x > 1$ , satisfying the equation  $\int_1^x t \ln t dt = \frac{1}{4}$  is

- A.  $\sqrt{e}$                                       B.  $e$                                       C.  $e^2$                                       D.  $e - 1$

gatecse2025-set2 calculus definite-integral one-mark

Answer key

### 5.2.4 Definite Integral: GATE CSE 2026 | Set 2 | Question: 17



For a real number  $a$ , let  $I(a) = \int_{-1}^1 (3x^2 - ax + 1) dx$ . Which of the following statements is/are true?

- A. The value of  $I(a)$  is independent of the value of  $a$   
 B. The value of  $I(a)$  can vary with the value of  $a$   
 C. There exists  $a \in (-\infty, +\infty)$  such that  $I(a)$  is a positive real number  
 D. There exists  $a \in (-\infty, +\infty)$  such that  $I(a)$  is a negative real number

gatecse-2026-set2 calculus definite-integral multiple-selects one-mark

Answer key

## 5.3 Differentiation (11)

Practice Tests: [Test 1 \(15Q\)](#) [Test 2 \(10Q\)](#)

### 5.3.1 Differentiation: GATE CSE 1996 | Question: 1.6



The formula used to compute an approximation for the second derivative of a function  $f$  at a point  $x_0$  is

- A.  $\frac{f(x_0 + h) + f(x_0 - h)}{2}$                                       B.  $\frac{f(x_0 + h) - f(x_0 - h)}{2h}$   
 C.  $\frac{f(x_0 + h) + 2f(x_0) + f(x_0 - h)}{h^2}$                                       D.  $\frac{f(x_0 + h) - 2f(x_0) + f(x_0 - h)}{h^2}$

gate1996 calculus differentiation normal

Answer key

5.3.2 Differentiation: GATE CSE 2014 | Set 1 | Question: 46



The function  $f(x) = x \sin x$  satisfies the following equation:

$$f''(x) + f(x) + t \cos x = 0$$

The value of  $t$  is \_\_\_\_\_.

gatecse-2014-set1 calculus easy numerical-answers differentiation

Answer key

5.3.3 Differentiation: GATE CSE 2014 | Set 1 | Question: 6



Let the function

$$f(\theta) = \begin{vmatrix} \sin \theta & \cos \theta & \tan \theta \\ \sin(\frac{\pi}{6}) & \cos(\frac{\pi}{6}) & \tan(\frac{\pi}{6}) \\ \sin(\frac{\pi}{3}) & \cos(\frac{\pi}{3}) & \tan(\frac{\pi}{3}) \end{vmatrix}$$

where

$\theta \in [\frac{\pi}{6}, \frac{\pi}{3}]$  and  $f'(\theta)$  denote the derivative of  $f$  with respect to  $\theta$ . Which of the following statements is/are **TRUE**?

- I. There exists  $\theta \in (\frac{\pi}{6}, \frac{\pi}{3})$  such that  $f'(\theta) = 0$
- II. There exists  $\theta \in (\frac{\pi}{6}, \frac{\pi}{3})$  such that  $f'(\theta) \neq 0$

- A. I only                      B. II only                      C. Both I and II                      D. Neither I nor II

gatecse-2014-set1 calculus differentiation normal

Answer key

5.3.4 Differentiation: GATE CSE 2016 | Set 2 | Question: 02



Let  $f(x)$  be a polynomial and  $g(x) = f'(x)$  be its derivative. If the degree of  $(f(x) + f(-x))$  is 10, then the degree of  $(g(x) - g(-x))$  is \_\_\_\_\_.

gatecse-2016-set2 calculus normal numerical-answers differentiation

Answer key

5.3.5 Differentiation: GATE CSE 2017 | Set 2 | Question: 10



If  $f(x) = R \sin(\frac{\pi x}{2}) + S$ ,  $f'(\frac{1}{2}) = \sqrt{2}$  and  $\int_0^1 f(x) dx = \frac{2R}{\pi}$ , then the constants  $R$  and  $S$  are

- A.  $\frac{2}{\pi}$  and  $\frac{16}{\pi}$                       B.  $\frac{2}{\pi}$  and 0                      C.  $\frac{4}{\pi}$  and 0                      D.  $\frac{4}{\pi}$  and  $\frac{16}{\pi}$

gatecse-2017-set2 engineering-mathematics calculus differentiation

Answer key

5.3.6 Differentiation: GATE CSE 2024 | Set 1 | Question: 1



Let  $f: \mathbb{R} \rightarrow \mathbb{R}$  be a function such that  $f(x) = \max\{x, x^3\}$ ,  $x \in \mathbb{R}$ , where  $\mathbb{R}$  is the set of all real numbers. The set of all points where  $f(x)$  is NOT differentiable is

- A.  $\{-1, 1, 2\}$                       B.  $\{-2, -1, 1\}$                       C.  $\{0, 1\}$                       D.  $\{-1, 0, 1\}$

gatecse-2024-set1 calculus differentiation one-mark

Answer key

### 5.3.7 Differentiation: GATE CSE 2025 | Set 1 | Question: 21



Consider the given function  $f(x)$ .

$$f(x) = \begin{cases} ax + b & \text{for } x < 1 \\ x^3 + x^2 + 1 & \text{for } x \geq 1 \end{cases}$$

If the function is differentiable everywhere, the value of  $b$  must be \_\_\_\_\_. (rounded off to one decimal place)

gatecse2025-set1 calculus differentiation numerical-answers one-mark

Answer key

### 5.3.8 Differentiation: GATE CSE 2026 | Set 1 | Question: 36



Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be defined as follows:

$$f(x) = \left( \frac{|x|}{2} - x \right) \left( x - \frac{|x|}{2} \right)$$

Which of the following statements is/are true?

- A.  $f$  has a local maximum
- B.  $f$  has a local minimum
- C.  $f'$  is continuous over  $\mathbb{R}$
- D.  $f'$  is not differentiable over  $\mathbb{R}$

gatecse-2026-set1 two-marks differentiation calculus multiple-selects

Answer key

### 5.3.9 Differentiation: GATE DA 2025 | Question: 14



Consider two functions  $f : \mathbb{R} \rightarrow \mathbb{R}$  and  $g : \mathbb{R} \rightarrow (1, \infty)$ . Both functions are differentiable at a point  $c$ . Which of the following functions is/are ALWAYS differentiable at  $c$ ? The symbol  $\cdot$  denotes product and the symbol  $\circ$  denotes composition of functions.

- A.  $f \pm g$
- B.  $f \cdot g$
- C.  $\frac{f}{g}$
- D.  $f \circ g + g \circ f$

gateda-2025 calculus differentiation limits multiple-selects one-mark

Answer key

### 5.3.10 Differentiation: GATE DA 2025 | Question: 4



Let  $f(x) = \frac{e^x - e^{-x}}{2}$ ,  $x \in \mathbb{R}$ . Let  $f^{(k)}(a)$  denote the  $k^{\text{th}}$  derivative of  $f$  evaluated at  $a$ . What is the value of  $f^{(10)}(0)$ ? (Note:  $!$  denotes factorial)

- A. 0
- B. 1
- C.  $\frac{1}{10!}$
- D.  $\frac{2}{10!}$

gateda-2025 calculus differentiation one-mark

Answer key

### 5.3.11 Differentiation: GATE DA 2025 | Question: 41



Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be a twice-differentiable function and suppose its second derivative satisfies  $f''(x) > 0$  for all  $x \in \mathbb{R}$ . Which of the following statements is/are ALWAYS correct?

- A.  $f$  has a local minima
- B. There does not exist  $x$  and  $y$ ,  $x \neq y$ , such that  $f'(x) = f'(y) = 0$
- C.  $f$  has at most one global minimum
- D.  $f$  has at most one local minimum

gateda-2025 calculus maxima-minima differentiation multiple-selects two-marks

Answer key

 **Practice Test:** Test 1 (14Q)

### 5.4.1 Integration: GATE CSE 1993 | Question: 02.6

The value of the double integral  $\int_0^1 \int_0^{\frac{1}{x}} \frac{x}{1+y^2} dx dy$  is\_\_\_\_\_.

gate1993 calculus integration normal fill-in-the-blanks

Answer key 

#### 5.4.2 Integration: GATE CSE 1998 | Question: 8

a. Find the points of local maxima and minima, if any, of the following function defined in  $0 \leq x \leq 6$ .

$$x^3 - 6x^2 + 9x + 15$$

b. Integrate

$$\int_{-\pi}^{\pi} x \cos x dx$$

gate1998 calculus maxima-minima integration normal descriptive

Answer key 



### 5.4.3 Integration: GATE CSE 2000 | Question: 2.3

Let  $S = \sum_{i=3}^{100} i \log_2 i$ , and  $T = \int_2^{100} x \log_2 x dx$ .

Which of the following statements is true?

- A.  $S > T$   
 B.  $S = T$   
 C.  $S < T$  and  $2S > T$   
 D.  $2S \leq T$

gatecse-2000 calculus integration normal

Answer key 

#### 5.4.4 Integration: GATE CSE 2009 | Question: 25

$$\int_0^{\pi/4} (1 - \tan x)/(1 + \tan x) dx$$

- A. 0                      B. 1                      C.  $\ln 2$                       D.  $1/2\ln 2$

gatecse-2009 calculus integration normal

Answer key 



## 5.4.5 Integration: GATE CSE 2011 | Question: 31

Given  $i = \sqrt{-1}$ , what will be the evaluation of the definite integral  $\int_0^{\pi/2} \frac{\cos x + i \sin x}{\cos x - i \sin x} dx$  ?

- A. 0                      B. 2                      C.  $-i$                       D.  $i$

gatecse-2011 calculus integration normal

Answer key 



## 5.4.6 Integration: GATE CSE 2014 | Set 3 | Question: 47

The value of the integral given below is



$$\int_0^{\pi} x^2 \cos x \, dx$$

A.  $-2\pi$

B.  $\pi$

C.  $-\pi$

D.  $2\pi$

gatecse-2014-set3 calculus limits integration normal

Answer key

#### 5.4.7 Integration: GATE CSE 2014 | Set 3 | Question: 6



If  $\int_0^{2\pi} |x \sin x| dx = k\pi$ , then the value of  $k$  is equal to \_\_\_\_\_.

gatecse-2014-set3 calculus integration limits numerical-answers easy

Answer key

#### 5.4.8 Integration: GATE CSE 2015 | Set 1 | Question: 44



Compute the value of:

$$\int_{\frac{1}{\pi}}^{\frac{2}{\pi}} \frac{\cos(1/x)}{x^2} dx$$

gatecse-2015-set1 calculus integration normal numerical-answers

Answer key

#### 5.4.9 Integration: GATE CSE 2015 | Set 3 | Question: 45



If for non-zero  $x$ ,  $af(x) + bf(\frac{1}{x}) = \frac{1}{x} - 25$  where  $a \neq b$  then  $\int_1^2 f(x) dx$  is

A.  $\frac{1}{a^2 - b^2} \left[ a(\ln 2 - 25) + \frac{47b}{2} \right]$   
 C.  $\frac{1}{a^2 - b^2} \left[ a(2 \ln 2 - 25) + \frac{47b}{2} \right]$

B.  $\frac{1}{a^2 - b^2} \left[ a(2 \ln 2 - 25) - \frac{47b}{2} \right]$   
 D.  $\frac{1}{a^2 - b^2} \left[ a(\ln 2 - 25) - \frac{47b}{2} \right]$

gatecse-2015-set3 calculus integration normal

Answer key

#### 5.4.10 Integration: GATE CSE 2018 | Question: 16



The value of  $\int_0^{\pi/4} x \cos(x^2) dx$  correct to three decimal places (assuming that  $\pi = 3.14$ ) is \_\_\_\_\_

gatecse-2018 calculus integration normal numerical-answers one-mark

Answer key

#### 5.4.11 Integration: GATE IT 2005 | Question: 35



What is the value of  $\int_0^{2\pi} (x - \pi)^2 (\sin x) dx$

A.  $-1$

B.  $0$

C.  $1$

D.  $\pi$

gateit-2005 calculus integration normal

Answer key

### 5.5 Limits (15)

Practice Tests: [Test 1 \(15Q\)](#) [Test 2 \(15Q\)](#) [Test 3 \(8Q\)](#)

## 5.5.1 Limits: GATE CSE 1993 | Question: 02.1



$$\lim_{x \rightarrow 0} \frac{x(e^x - 1) + 2(\cos x - 1)}{x(1 - \cos x)} \text{ is } \underline{\hspace{2cm}}$$

gate1993 limits calculus normal fill-in-the-blanks

Answer key

## 5.5.2 Limits: GATE CSE 1995 | Question: 7(B)



Compute without using power series expansion  $\lim_{x \rightarrow 0} \frac{\sin x}{x}$ .

gate1995 calculus limits numerical-answers

Answer key

## 5.5.3 Limits: GATE CSE 2008 | Question: 1



$$\lim_{x \rightarrow \infty} \frac{x - \sin x}{x + \cos x} \text{ equals}$$

- A. 1                                      B. -1                                      C.  $\infty$                                       D.  $-\infty$

gatecse-2008 calculus limits easy

Answer key

## 5.5.4 Limits: GATE CSE 2010 | Question: 5



What is the value of  $\lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right)^{2n}$  ?

- A. 0                                      B.  $e^{-2}$                                       C.  $e^{-1/2}$                                       D. 1

gatecse-2010 calculus limits normal

Answer key

## 5.5.5 Limits: GATE CSE 2015 | Set 1 | Question: 4



$$\lim_{x \rightarrow \infty} x^{\frac{1}{x}} \text{ is}$$

- A.  $\infty$                                       B. 0                                      C. 1                                      D. Not defined

gatecse-2015-set1 calculus limits normal

Answer key

## 5.5.6 Limits: GATE CSE 2015 | Set 3 | Question: 9



The value of  $\lim_{x \rightarrow \infty} (1 + x^2)^{e^{-x}}$  is

- A. 0                                      B.  $\frac{1}{2}$                                       C. 1                                      D.  $\infty$

gatecse-2015-set3 calculus limits normal

Answer key

## 5.5.7 Limits: GATE CSE 2016 | Set 1 | Question: 3



$$\lim_{x \rightarrow 4} \frac{\sin(x - 4)}{x - 4} = \underline{\hspace{2cm}}$$

gatecse-2016-set1 calculus limits easy numerical-answers

Answer key

5.5.8 Limits: GATE CSE 2017 | Set 1 | Question: 28



The value of  $\lim_{x \rightarrow 1} \frac{x^7 - 2x^5 + 1}{x^3 - 3x^2 + 2}$

- A. is 0                      B. is  $-1$                       C. is 1                      D. does not exist

gatecse-2017-set1   calculus   limits   normal

Answer key

5.5.9 Limits: GATE CSE 2019 | Question: 13



Compute  $\lim_{x \rightarrow 3} \frac{x^4 - 81}{2x^2 - 5x - 3}$

- A. 1                                      B.  $53/12$   
C.  $108/7$                               D. Limit does not exist

gatecse-2019   engineering-mathematics   calculus   limits   one-mark

Answer key

5.5.10 Limits: GATE CSE 2021 | Set 1 | Question: 20



Consider the following expression.

$$\lim_{x \rightarrow -3} \frac{\sqrt{2x+22} - 4}{x+3}$$

The value of the above expression (rounded to 2 decimal places) is \_\_\_\_\_.

gatecse-2021-set1   calculus   limits   numerical-answers   one-mark

Answer key

5.5.11 Limits: GATE CSE 2022 | Question: 24



The value of the following limit is \_\_\_\_\_.

$$\lim_{x \rightarrow 0^+} \frac{\sqrt{x}}{1 - e^{2\sqrt{x}}}$$

gatecse-2022   numerical-answers   calculus   limits   one-mark

Answer key

5.5.12 Limits: GATE DA 2025 | Question: 22



$$\lim_{t \rightarrow +\infty} \sqrt{t^2 + t} - t =$$

(Round off to one decimal place)

gateda-2025   calculus   limits   numerical-answers   one-mark

Answer key

5.5.13 Limits: GATE DA 2025 | Question: 49



Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be such that  $|f(x) - f(y)| \leq (x - y)^2$  for all  $x, y \in \mathbb{R}$ . Then  $f(1) - f(0) =$  \_\_\_\_\_  
(Answer in integer)

Answer key

## 5.5.14 Limits: GATE DS&amp;AI 2024 | Question: 50

Evaluate the following limit:

$$\lim_{x \rightarrow 0} \frac{\ln((x^2 + 1) \cos x)}{x^2} =$$

Answer key

## 5.5.15 Limits: GATE Data Science and Artificial Intelligence 2024 | Sample Paper | Question: 5

$$\lim_{x \rightarrow 2} \frac{\sqrt{x} - \sqrt{2}}{x - 2}$$

A. 0

B.  $\sqrt{2}$ C.  $\frac{1}{2\sqrt{2}}$ D.  $\frac{1}{\sqrt{2}}$ 

Answer key

## 5.6

## Maxima Minima (14)

Practice Tests: Test 1 (15Q) Test 2 (15Q) Test 3 (6Q)

## 5.6.1 Maxima Minima: GATE CSE 1987 | Question: 1-xxvi

If  $f(x_i) \cdot f(x_{i+1}) < 0$  then

- A. There must be a root of  $f(x)$  between  $x_i$  and  $x_{i+1}$
- B. There need not be a root of  $f(x)$  between  $x_i$  and  $x_{i+1}$
- C. The fourth derivative of  $f(x)$  with respect to  $x$  vanishes at  $x_i$
- D. The fourth derivative of  $f(x)$  with respect to  $x$  vanishes at  $x_{i+1}$

Answer key

## 5.6.2 Maxima Minima: GATE CSE 1995 | Question: 1.21

In the interval  $[0, \pi]$  the equation  $x = \cos x$  has

- A. No solution
- B. Exactly one solution
- C. Exactly two solutions
- D. An infinite number of solutions

Answer key

## 5.6.3 Maxima Minima: GATE CSE 1995 | Question: 25a

Find the minimum value of  $3 - 4x + 2x^2$ .

Answer key

## 5.6.4 Maxima Minima: GATE CSE 1997 | Question: 4.1

What is the maximum value of the function  $f(x) = 2x^2 - 2x + 6$  in the interval  $[0, 2]$ ?

A. 6

B. 10

C. 12

D. 5.5

gate1997 calculus maxima-minima normal

Answer key

## 5.6.5 Maxima Minima: GATE CSE 2008 | Question: 25



A point on a curve is said to be an extremum if it is a local minimum or a local maximum. The number of distinct extrema for the curve  $3x^4 - 16x^3 + 24x^2 + 37$  is

A. 0

B. 1

C. 2

D. 3

gatecse-2008 calculus maxima-minima easy

Answer key

## 5.6.6 Maxima Minima: GATE CSE 2012 | Question: 9



Consider the function  $f(x) = \sin(x)$  in the interval  $x = \left[\frac{\pi}{4}, \frac{7\pi}{4}\right]$ . The number and location(s) of the local minima of this function are

A. One, at  $\frac{\pi}{2}$ B. One, at  $\frac{3\pi}{2}$ C. Two, at  $\frac{\pi}{2}$  and  $\frac{3\pi}{2}$ D. Two, at  $\frac{\pi}{4}$  and  $\frac{3\pi}{2}$ 

gatecse-2012 calculus maxima-minima normal

Answer key

## 5.6.7 Maxima Minima: GATE CSE 2015 | Set 2 | GA | Question: 3



Consider a function  $f(x) = 1 - |x|$  on  $-1 \leq x \leq 1$ . The value of  $x$  at which the function attains a maximum, and the maximum value of the function are:

A. 0, -1

B. -1, 0

C. 0, 1

D. -1, 2

gatecse-2015-set2 set-theory&amp;algebra functions normal maxima-minima

Answer key

## 5.6.8 Maxima Minima: GATE CSE 2020 | Question: 1



Consider the functions

I.  $e^{-x}$ II.  $x^2 - \sin x$ III.  $\sqrt{x^3 + 1}$ 

Which of the above functions is/are increasing everywhere in  $[0, 1]$ ?

A. III only

B. II only

C. II and III only

D. I and III only

gatecse-2020 engineering-mathematics calculus maxima-minima one-mark

Answer key

## 5.6.9 Maxima Minima: GATE CSE 2023 | Question: 18



Let

$$f(x) = x^3 + 15x^2 - 33x - 36$$

be a real-valued function.

Which of the following statements is/are TRUE?

A.  $f(x)$  does not have a local maximum.

B.  $f(x)$  has a local maximum.

C.  $f(x)$  does not have a local minimum.

D.  $f(x)$  has a local minimum.

gatecse-2023 calculus maxima-minima multiple-selects one-mark

Answer key

#### 5.6.10 Maxima Minima: GATE DA 2025 | Question: 39



Consider the function  $f(x) = \frac{x^3}{3} + \frac{7}{2}x^2 + 10x + \frac{133}{2}$ ,  $x \in [-8, 0]$ . Which of the following statements is/are correct?

A. The maximum value of  $f$  is attained at  $x = -5$

B. The minimum value of  $f$  is attained at  $x = -2$

C. The maximum value of  $f$  is  $\frac{133}{2}$

D. The minimum value of the derivative of  $f$  is attained at  $x = -\frac{7}{2}$

gateda-2025 calculus maxima-minima multiple-selects two-marks

Answer key

#### 5.6.11 Maxima Minima: GATE DA 2026 | Question: 17



Let  $f(x) = x^3 - 3x^2 + 2$  be a function defined on  $(-1, 3]$ .

Which of the following statements is/are correct?

A.  $f(x)$  has exactly two roots in  $[-0.9, 0]$ .

B.  $f(x)$  has a minimum at 2 only.

C.  $f(x)$  has maximum at 0 only.

D.  $f(x)$  has a root at 1.

gateda-2026 calculus maxima-minima multiple-selects one-mark

Answer key

#### 5.6.12 Maxima Minima: GATE DS&AI 2024 | Question: 40



Consider the function  $f : \mathbb{R} \rightarrow \mathbb{R}$  where  $\mathbb{R}$  is the set of all real numbers.

$$f(x) = \frac{x^4}{4} - \frac{2x^3}{3} - \frac{3x^2}{2} + 1$$

Which of the following statements is/are TRUE?

A.  $x = 0$  is a local maximum of  $f$

B.  $x = 3$  is a local minimum of  $f$

C.  $x = -1$  is a local maximum of  $f$

D.  $x = 0$  is a local minimum of  $f$

gate-ds-ai-2024 calculus maxima-minima multiple-selects two-marks

Answer key

#### 5.6.13 Maxima Minima: GATE DS&AI 2024 | Question: 5



For any twice differentiable function  $f : \mathbb{R} \rightarrow \mathbb{R}$ , if at some  $x^* \in \mathbb{R}$ ,  $f'(x^*) = 0$  and  $f''(x^*) > 0$ , then the function  $f$  necessarily has a \_\_\_\_\_ at  $x = x^*$ .

Note:  $\mathbb{R}$  denotes the set of real numbers.

A. local minimum

B. global minimum

C. local maximum

D. global maximum

gate-ds-ai-2024 calculus maxima-minima one-mark

Answer key

### 5.6.14 Maxima Minima: GATE IT 2008 | Question: 31



If  $f(x)$  is defined as follows, what is the minimum value of  $f(x)$  for  $x \in (0, 2]$  ?

$$f(x) = \begin{cases} \frac{25}{8x} & \text{when } x \leq \frac{3}{2} \\ x + \frac{1}{x} & \text{otherwise} \end{cases}$$

A. 2

B.  $2\frac{1}{12}$

C.  $2\frac{1}{6}$

D.  $2\frac{1}{2}$

gateit-2008 calculus maxima-minima normal

Answer key

## 5.7

### Polynomials (2)

#### 5.7.1 Polynomials: GATE CSE 1987 | Question: 1-xxii



The equation  $7x^7 + 14x^6 + 12x^5 + 3x^4 + 12x^3 + 10x^2 + 5x + 7 = 0$  has

A. All complex roots

B. At least one real root

C. Four pairs of imaginary roots

D. None of the above

gate1987 calculus polynomials

Answer key

#### 5.7.2 Polynomials: GATE CSE 1995 | Question: 2.8



If the cube roots of unity are  $1, \omega$  and  $\omega^2$ , then the roots of the following equation are

$$(x - 1)^3 + 8 = 0$$

A.  $-1, 1 + 2\omega, 1 + 2\omega^2$

B.  $1, 1 - 2\omega, 1 - 2\omega^2$

C.  $-1, 1 - 2\omega, 1 - 2\omega^2$

D.  $-1, 1 + 2\omega, -1 + 2\omega^2$

gate1995 calculus normal polynomials

Answer key

## 5.8

### Summation (1)

#### 5.8.1 Summation: GATE DA 2026 | Question: 25



The value of  $\sum_{i=0}^{\infty} \sum_{j=1}^{\infty} 2^{-i} 3^{-j}$  is \_\_\_\_\_. (Answer in integer)

gateda-2026 calculus numerical-answers one-mark summation

Answer key

## Answer Keys

|        |               |        |             |        |         |        |         |        |     |
|--------|---------------|--------|-------------|--------|---------|--------|---------|--------|-----|
| 5.1.1  | N/A           | 5.1.2  | B           | 5.1.3  | A       | 5.1.4  | A       | 5.1.5  | A   |
| 5.1.6  | C             | 5.1.7  | 19 : 19     | 5.1.8  | 03 : 03 | 5.1.9  | 40:40   | 5.1.10 | A   |
| 5.1.11 | C             | 5.2.1  | 0           | 5.2.2  | B       | 5.2.3  | A       | 5.2.4  | A;C |
| 5.3.1  | D             | 5.3.2  | -2          | 5.3.3  | C       | 5.3.4  | 9       | 5.3.5  | C   |
| 5.3.6  | D             | 5.3.7  | -2.1:-1.9   | 5.3.8  | A;C;D   | 5.3.9  | A;B;C   | 5.3.10 | A   |
| 5.3.11 | B;C;D         | 5.4.1  | N/A         | 5.4.2  | N/A     | 5.4.3  | A       | 5.4.4  | D   |
| 5.4.5  | D             | 5.4.6  | A           | 5.4.7  | 4       | 5.4.8  | -1      | 5.4.9  | A   |
| 5.4.10 | 0.288 : 0.289 | 5.4.11 | B           | 5.5.1  | 1       | 5.5.2  | 1       | 5.5.3  | A   |
| 5.5.4  | B             | 5.5.5  | C           | 5.5.6  | C       | 5.5.7  | 1       | 5.5.8  | C   |
| 5.5.9  | C             | 5.5.10 | 0.25 : 0.25 | 5.5.11 | -0.5    | 5.5.12 | 0.5:0.5 | 5.5.13 | 0:0 |

|        |     |
|--------|-----|
| 5.5.14 | 0.5 |
| 5.6.4  | B   |
| 5.6.9  | B;D |
| 5.6.14 | B   |

|        |     |
|--------|-----|
| 5.5.15 | C   |
| 5.6.5  | B   |
| 5.6.10 | C;D |
| 5.7.1  | B   |

|        |     |
|--------|-----|
| 5.6.1  | A   |
| 5.6.6  | D   |
| 5.6.11 | B;D |
| 5.7.2  | C   |

|        |     |
|--------|-----|
| 5.6.2  | B   |
| 5.6.7  | C   |
| 5.6.12 | A;B |
| 5.8.1  | 1:1 |

|        |   |
|--------|---|
| 5.6.3  | 1 |
| 5.6.8  | A |
| 5.6.13 | A |